

Measuring current



Measuring resistance



Measuring voltage

A multimeter is a very useful electronic instrument used to measure voltage, current, resistance and capacitance. In fact, it can also measure AC voltage and AC current. The circular slider has to be kept in appropriate position to measure each electrical quantity.

2.2.4 Temperature dependence of resistivity

The resistivity of a material is dependent on temperature. It is experimentally found that for a wide range of temperatures, the resistivity of a conductor increases with increase in temperature according to the expression,

$$\rho_T = \rho_\circ\left[1+\alpha\left(T-T_\circ\right)\right] \qquad (2.27)$$

where ρ_T is the resistivity of a conductor at $T\,^\circ C$, ρ_o is the resistivity of the conductor at some reference temperature T_o (usually at 20°C) and α is the temperature coefficient of resistivity. **It is defined as the ratio of increase in resistivity per degree rise in temperature to its resistivity at T_o.**

From the equation (2.27), we can write

$$\rho_T - \rho_\circ = \alpha\rho_\circ\left(T-T_\circ\right)$$

$$\therefore \alpha = \frac{\rho_T-\rho_\circ}{\rho_\circ\left(T-T_\circ\right)} = \frac{\Delta\rho}{\rho_\circ\Delta T}$$

where $\Delta\rho = \rho_T - \rho_o$ is change in resistivity for a change in temperature $\Delta T = T - T_o$. Its unit is per °C.

α of conductors

For conductors α *is positive.* If the temperature of a conductor increases, the average kinetic energy of electrons in the conductor increases. This results in more frequent collisions and hence the resistivity increases. The graph of the equation (2.27) is shown in Figure 2.13

Even though, the resistivity of conductors like metals varies linearly for wide range of temperatures, there also exists a non-linear region at very low temperatures. The resistivity approaches some finite value as the temperature approaches absolute zero as shown in Figure 2.13(b).

As the resistance is directly proportional to resistivity of the material, we can also write the resistance of a conductor at temperature T °C as

$$R_T = R_\circ\left[1+\alpha\left(T-T_\circ\right)\right] \qquad (2.28)$$

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